A Theoretical Analysis of the Test Error of Finite-Rank Kernel Ridge Regression

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Contribution

1. Precise non-asymptotic upper and lower bounds for test error;
2. with ridge λ decoupled from sample size N;
3. including inconsistent case (target function outside RKHS)

for under-parametrized kernel ridge regression.

Kernel Ridge Regression

A (finite) rank-M kernel K is a map $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ defined for any $x, x' \in \mathcal{X}$ by

$$K(x, x') = \sum_{k=1}^{M} \lambda_k \psi_k(x) \psi_k(x'), \qquad (1)$$

where the positive numbers $\{\lambda_k\}_{k=1}^{M}$ are called the eigenvalues, and orthonormal functions $\psi_k \in L^2_\rho$ are called the eigenfunctions. Fix a ridge $\lambda > 0$, the regressor $f_{\mathbf{Z},\lambda}$ of the finite-rank kernel K is the (unique) minimizer of

$$f_{\mathbf{Z},\lambda} \stackrel{\text{def.}}{=} \min_{f \in \mathcal{H}} \frac{1}{N} \sum_{i=1}^{N} (f(x_i) - y_i)^2 + \lambda \|f\|^2_{\mathcal{H}}. \qquad (2)$$

where $\mathcal{H} \stackrel{\text{def.}}{=} \operatorname{span}\{\lambda_1^{-1/2}\psi_1, \ldots, \lambda_M^{-1/2}\psi_M\}$ the reproducing kernel Hilbert space (RKHS) associated to K, and $\mathbf{Z} = (\mathbf{X}, \mathbf{y})$ a sample of size N.

Test Error

- Note that $\{\psi_k\}_{k=1}^{M} \subset L^2_\rho$ are orthonormal functions;
- write target function $\tilde{f} = \sum_{k=1}^{M} \tilde{\gamma}_k \psi_k + \tilde{\gamma}_{>M} \psi_{>M}$, where
- $\tilde{\gamma}_k \in \mathbb{R}$, $\psi_{>M} \perp \psi_k$, $\forall k$;
- assume noisy sampling: $y_i = \tilde{f}(x_i) + \epsilon_i$ where
- $\mathbb{E}[\epsilon] = 0$, $\mathbb{E}[\epsilon^2] = \sigma^2 > 0$.

The finite-rank KRR (2)'s test error is

$$\mathcal{R}_{\mathbf{Z},\lambda} \stackrel{\text{def.}}{=} \mathbb{E}_{x,\epsilon}\left[(f_{\mathbf{Z},\lambda}(x) - \tilde{f}(x))^2\right]. \qquad (3)$$

Bias-Variance Decomposition:

$$\mathcal{R}_{\mathbf{Z},\lambda} = \text{bias} + \text{variance},$$

where bias measures the error when there is no noise in the label, and variance is defined to be the difference between test error and bias: variance $\stackrel{\text{def.}}{=} \mathcal{R}_{\mathbf{Z},\lambda} - \text{bias}$.

References

[1] Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar: Foundations of machine learning, 340 MIT press (2018)

[2] Arash Amini, Richard Baumgartner, and Dai Feng. Target alignment in truncated kernel ridge regression, Advances in Neural Information Processing Systems, 35:21948–21960 (2022)

[3] Francis Bach. Learning theory from first principles, https://www.di.ens.fr/~fbach/ltfp_book.pdf (2023)

Comparison

Assumptions / results	Mohri et al.	Amini et al.	Bach	This paper
Include inconsistent case	✓	✗	(✓)*	✓
Bias-variance decomposition	✗	✓	✓	✓
Test error high probability upper bound	✓	✗	✓	✓
Test error high probability lower bound	✗	✗	✗	✓
Bounds improve with smaller ridge?	✗	-	✗	✓
Upper bound decay rate**	$\sqrt{\frac{\log N}{N}}$	-	$\lambda + \frac{\log N}{\lambda N}$	$(\lambda + \frac{1}{N})\sqrt{\frac{\log N}{N}}$

Comparison of risk-bounds for finite-rank kernel ridge regressors with previous works: Mohri et al. [1] gives a loose upper bound by Rademacher-complexity; Amini et al. [2] indeed gives an exact formula for the train error instead of the test error; Bach [3] gives an upper bound, which becomes loose when the ridge λ is small and the model is under-parametrized. (*): The bound for the inconsistent case is implicit. (**): By decay, we mean the difference between the upper bound and its limit as $N \to \infty$.

Main Result

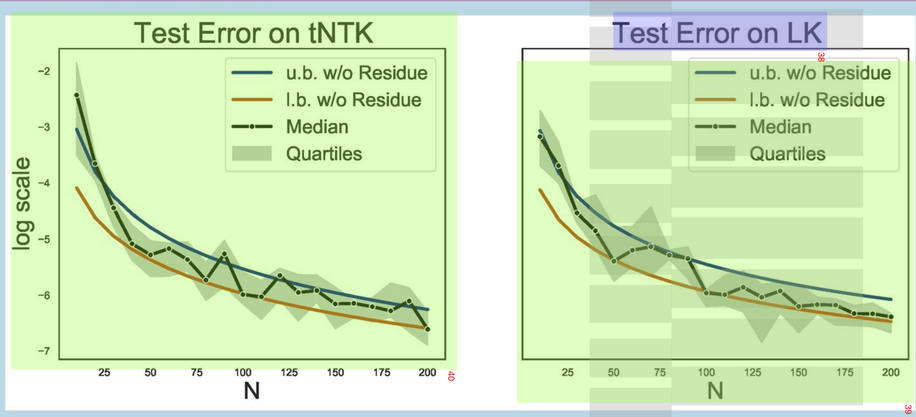
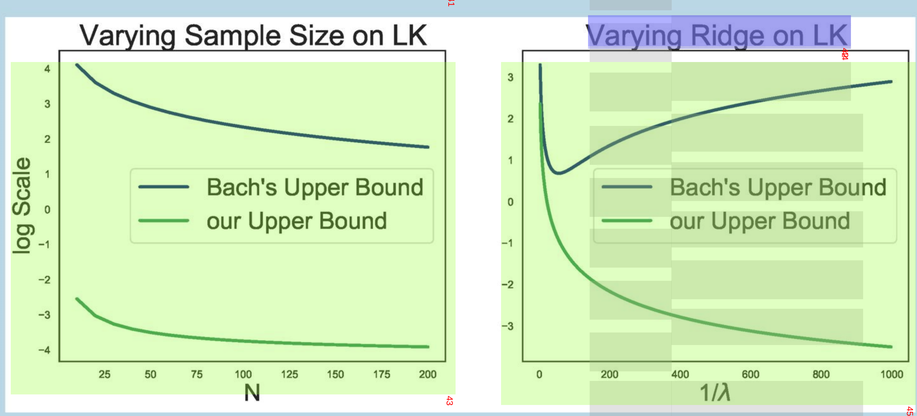
Under mild conditions on the kernel K, for N sufficiently large, there exist some constants C_1, C_2 independent to N and λ such that, with a probability of at least $1 - 2/N$ w.r.t. random sampling, we have the bounds on the test error $\mathcal{R}_{\mathbf{Z},\lambda}$:

$$\mathcal{R}_{\mathbf{Z},\lambda} \le \tilde{\gamma}^2_{>M} + \lambda \|\tilde{f}_{\le M}\|^2_{\mathcal{H}} + \left(\frac{1}{4}\|\tilde{f}\|^2_{L^2_\rho} + 2\lambda \|\tilde{f}_{\le M}\|^2_{\mathcal{H}}\right)\sqrt{\frac{\log N}{N}} + C_1 \frac{\log N}{N}$$
$$+ \sigma^2 \frac{M}{N}\left(1 + \sqrt{\frac{\log N}{N}} + C_2 \frac{\log N}{N}\right);$$

$$\mathcal{R}_{\mathbf{Z},\lambda} \ge \tilde{\gamma}^2_{>M} + \frac{\lambda^2 \lambda_M}{(\lambda_k + \lambda)^2} \|\tilde{f}_{\le M}\|^2_{\mathcal{H}} - \left(\frac{1}{4}\|\tilde{f}\|^2_{L^2_\rho} + \frac{2\lambda^2}{\lambda_1 + \lambda}\|\tilde{f}_{\le M}\|^2_{\mathcal{H}}\right)\sqrt{\frac{\log N}{N}} - C_1 \frac{\log N}{N}$$
$$+ \sigma^2 \frac{M}{N}\left(1 - \sqrt{\frac{\log N}{N}} - C_2 \frac{\log N}{N}\right).$$

Experiment

Our bounds are sharp and capture the decay of the test error in N: the figure below shows the simulation on two different kernels:



We also compare our upper bound with Bach [3]: